JUILLIARD ACADEMY

Interhouse Sports Day — Product Requirement Document (PRD)

PLATFORM IDENTIFIER	TARGET SESSION	DOCUMENT STATUS	VERSION
ja-mega-26	2026 Academic Year	Production Baseline	1.0.0 (Official Release)

Platform Identifier: `ja-mega-26`
Application Title: Juilliard Academy Interhouse Sports Day Web Platform
Target Release: 2026 Academic Session
Document Status: Production Baseline
Document Version: 1.0.0
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1. Executive Summary & Product Vision

1.1 Overview

The **Juilliard Academy Interhouse Sports Day Application** is a real-time tournament broadcasting and official scoring platform purpose-built for Juilliard Academy's premier athletic competitions. Inspired by world-class professional sports platforms such as **FotMob**, **Sofascore**, and **ESPN**, the application bridges high-octane stadium excitement with precision sports administrative governance.

The platform provides an end-to-end digital experience:

- **For Spectators, Students, and Parents:** Sub-second live scoreboards, match fixtures, interactive 3D podiums, athlete bios, high-resolution multimedia gallery, video highlights, and social engagement (likes, comments, and hashtag streams).
- **For Officials, Scrutineers, and Judges:** A secure, role-gated administrative console for live timing entry, race placings, match statistics, football and basketball knockout brackets, student roster assignment, and immutable audit logging.

1.2 Mission Statement

> *"To deliver an immersive, lightning-fast, and transparent digital stadium atmosphere that elevates student athletic achievements, unites the school community behind their houses, and guarantees absolute integrity in competition scoring."*

1.3 Key Value Propositions

1. **FotMob-Caliber Match & Event Experience:** Professional match cards, live status pulses, event countdowns, tactical head-to-head records, and play-by-play timelines.
2. **Dual-Competition Architecture:** Seamless real-time switching between the flagship **Interhouse Sports Championship** (4 Houses) and the **Prefectship Cup** (4 Prefectorial Teams).

3. **Olympic-Standard Medal Tallying & Multi-Format Scoring:** Strict Olympic medal hierarchy (Gold > Silver > Bronze) for track and field, coupled with football group/knockout points tables and basketball knockout brackets.
4. **Master Student Roster & Athlete Profiles:** Full integration with a 1,000+ student directory featuring grades (JSS 1 through SSS 3), house rosters, personal bests, and career medal counts.
5. **Realtime Collaborative Official Console:** High-precision scoring interface with datalist auto-completion, tie-break handlers, penalty shootouts, and an automated audit trail for dispute prevention.
6. **Zero-Latency Client-Side Performance:** Built with lightweight Vanilla ES Modules, CSS variables, and Web Audio API procedural sound synthesis (0KB audio asset payload) backed by Google Cloud Firestore real-time websockets.

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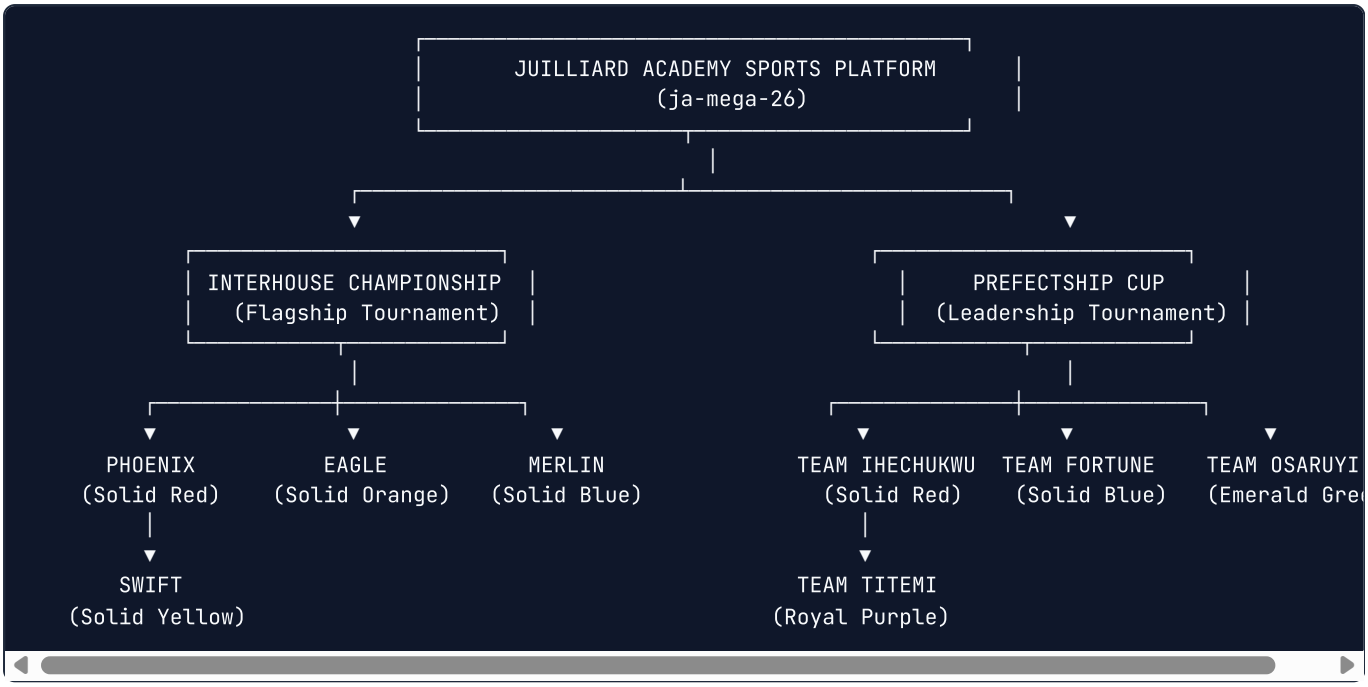
2. Stakeholders & User Personas

Persona	Primary Needs & Motivations	Typical Journey	Permissions
Student Athlete	Check scheduled heat times, monitor house standings, view personal profile, verify race times and medal records.	Arrives via mobile → selects house → navigates to **Matches** or **Table** → views individual profile via athlete search.	Public Read
Spectator / Parent	Follow live matches in real-time from the stands or remotely, watch video highlights, read match reports, cheer and comment.	Opens link → selects house affiliation → tracks live ticker → likes/comments on news stories.	Public Read + Comment / Like
House Master / Mistress	Track house performance across categories, inspect athlete rosters, cheer team members, strategize event participation.	Inspects **Houses** hub → reviews house roster and medal breakdown → monitors live leaderboard.	Public Read
Judge / Scrutineer / Official	Input track sprint times, field jump distances, head-to-head ball scores, award podium finishes, manage official notices.	Logs in via PIN / Google Auth → opens **Judge Console** → enters score modal → publishes official results live.	Admin Write (`admin`)
Super Administrator / Head Patron	Maintain system governance, audit all scoring changes, manage official accounts, conduct batch student transfers, emergency resets.	Accesses **Admin Console** → audits real-time logs → creates/manages official credentials → oversees full championship data.	Super Admin (`super_admin`)

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3. Domain Model & Tournament Architecture

The platform supports a **Dual-Tournament Architecture** allowing users to switch between tournaments on the fly while maintaining independent data caches, state isolation, and visual branding.



3.1 Interhouse Championship

The flagship inter-house sporting competition comprising the four historic houses of Juilliard Academy:

House ID	House Name	Brand Color	Mascot	House Motto	Leadership & Patrons
`phoenix`	**Phoenix**	`#EF4444` (Solid Red)	Phoenix Bird	*"From the Ashes, We Rise to Glory"*	**Captain:** Chinedu Ihejiahi (SS 3) **Patrons:** Mrs. Angela Davies, Mr. Peter O'Connor, Ms. Fatima Bello
`eagle`	**Eagle**	`#F97316` (Solid Orange)	Eagle Bird	*"Soar High with Honor and Courage"*	**Captain:** Donald Ariyo (SS 3) **Patrons:** Mr. Harrison, Ms. Chen, Coach Adams
`merlin`	**Merlin**	`#2563EB` (Solid Blue)	Merlin Falcon	*"Wisdom, Power & Unyielding Focus"*	**Captain:** Michelle Olurotimi (SS 3) **Patrons:** Dr. Evelyn Wright, Mr. Kenneth Cole, Mrs. Sarah Jenkins
`swift`	**Swift**	`#FACC15` (Solid Yellow)	Swift Bird	*"Swift as Lightning, United as One"*	**Captain:** Adefilewa Adebimpe (SS 3) **Patrons:** Ms. Rebecca Taylor, Coach Ibrahim Bello, Mr. David Alabi

3.2 Prefectship Cup (Prefectorial Tournament)

A special championship contested by the senior student leadership council teams:

Team ID	Team Name	Brand Color	Mascot	Motto	Team Captain
`team-ihechukwu`	**Team Ihechukwu**	`#EF4444` (Solid Red)	Crown & Lion	*"Leadership, Valor & Victory"*	Ihechukwu (Senior Prefect)
`team-fortune`	**Team Fortune**	`#2563EB` (Solid Blue)	Falcon of Victory	*"Fortune Favors the Bold & Relentless"*	Fortune (Sports Prefect)
`team-osaruyi`	**Team Osaruyi**	`#10B981` (Emerald Green)	Emerald Stallion	*"Relentless Drive, Speed & Discipline"*	Osaruyi (Senior Prefect)
`team-titemi`	**Team Titemi**	`#8B5CF6` (Royal Purple)	Purple Lightning	*"Strength in Unity, Focus & Power"*	Titemi (Senior Prefect)

3.3 Dynamic Competition Switcher

- Accessible via global desktop header dropdown, dedicated route (`#/competitions` , `#/select-cup`), or mobile bottom navigation.
- Instantly toggles `localStorage["juilliard_active_competition"]` between `interhouse` and `prefectship` .
- Dynamically shifts UI color accents, crest assets, navigation branding, and switches Firestore subscription contexts without requiring a browser page reload.

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4. Competition Rules & Athletic Scoring Engine

The scoring engine implements strict athletic federation standards adapted for Juilliard Academy.

4.1 Olympic-Style Medal Standings Hierarchy

Overall championship rank is determined by medal superiority, not arbitrary point tallies:

1. **Gold Medal Count** (Highest priority)
2. **Silver Medal Count** (Tie-breaker 1)
3. **Bronze Medal Count** (Tie-breaker 2)
4. **Total Medals Count** (Tie-breaker 3)
5. **Team Points** (Tie-breaker 4: Football & Basketball league points)
6. **Fourth-Place Finishes** (Tie-breaker 5)

```
Ranking Criteria: Gold → Silver → Bronze → Total Medals → Team Points → 4th Finishes
```

4.2 Multi-House Events (Track, Field & Novelty)

- Formats: `format: "multi"`
- Positions & Medals:
  - **1st Place:** Gold Medal
  - **2nd Place:** Silver Medal
  - **3rd Place:** Bronze Medal
  - **4th Place:** 4th place finish recorded for tie-breaking
- Validation: Times (e.g. `10.42s`), distances (e.g. `14.85m`), and athlete names selected from the student registry.
- Points: General track and field events award zero arbitrary league points; rankings reflect pure Olympic medal tallies.

4.3 Head-to-Head Ball Sports (Football & Basketball)

- Formats: `format: "h2h"`

FOOTBALL (SOCCER)

- **Tournament Structure:** 3 Matchday Group Stage (each house plays 3 matches) + Knockout Final & 3rd Place Match.
- **Standings Matrix:** Matches Played (P), Won (W), Drawn (D), Lost (L), Goals For (GF), Goals Against (GA), Goal Difference (GD), Points (PTS).
- **Points Awarded:** Win = 3 points, Draw = 1 point, Loss = 0 points.
- **Medal Distribution:**
  - Group Stage Matches: Award league points only; **zero medals**.
  - Drawn Matches: Award 1 point each; **zero medals**.
  - Gold Medal Final: Winner = Gold Medal, Runner-Up = Silver Medal.
  - 3rd Place Playoff: Winner = Bronze Medal.

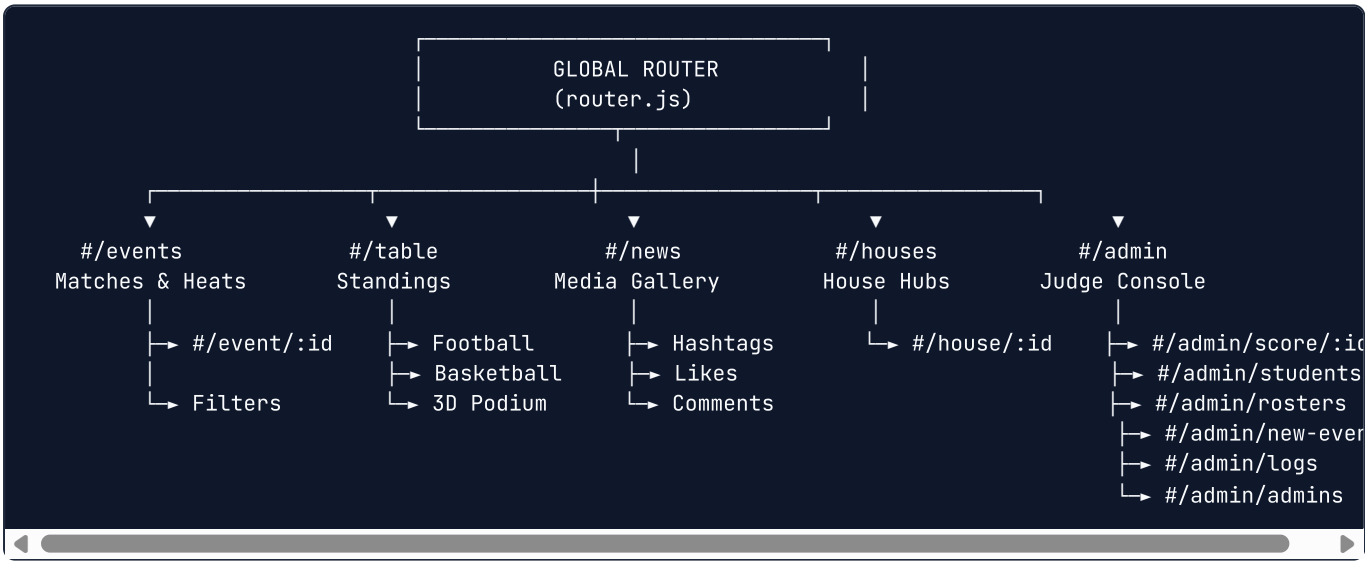
BASKETBALL

- **Tournament Structure:** Pure Knockout Bracket.
  - Semi-Final 1 & Semi-Final 2.
  - 3rd Place Playoff (Bronze Medal).
  - Grand Championship Final (Gold & Silver Medals).
- **Match Tracking:** Quarter scores, halftime scores, buzzer-beaters, total team points.

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5. Functional Requirements & Feature Specifications

5.1 Module Matrix & Navigation Map



5.2 Feature Specifications

FR-01: ONBOARDING & HOUSE SUPPORTER AFFILIATION

- **Routes:** `#/onboarding`, `#/welcome`
- **Functional Requirements:**
  - First-time visitors are presented with interactive house crest cards with glow animations.
  - Selecting a house or team saves the choice to `localStorage.juilliard_supported_house` and `localStorage.juilliard_supported_pref_team`.

- Automatically customizes the user experience (highlighting supported house fixtures, showing personalized cheer banners).
- Provides quick access to Official Portal login for scrutineers.

FR-02: MATCHES & FIXTURES SCHEDULE

- **Route:** `#/events`
- **Functional Requirements:**
  - FotMob-style fixtures layout organized chronologically or grouped by category.
  - **Category Tabs:** All, Track, Field, Football, Basketball, Novelty.
  - **Gender Filtering:** All, Boys, Girls, Mixed.
  - **Status Filtering:** All, Live, Upcoming, Completed.
  - **Match Cards:** Display contending houses, scheduled start time, live game clock or final score badge, venue lane, and status pills (`LIVE` pulsating red, `UPCOMING` amber, `COMPLETED` green).
  - Clicking any card transitions to the dedicated Event Detail screen (`#/event/:id`).

FR-03: EVENT & MATCH DETAIL VIEW

- **Route:** `#/event/:id`
- **Functional Requirements:**
  - Dynamic high-impact scoreboard header featuring house crests, background gradients, and live status indicator.
  - **Head-to-Head Ball Matches:**
    - Live score, match status (e.g. "FT", "HT", "Penalties").
    - Tabbed view: **Match Facts** (scorers, cards, stadium), **Live Commentary / Timeline**, **Lineups & Squads**, **Head-to-Head History**.
  - **Track & Field Events:**
    - Visual 1st, 2nd, 3rd podium finisher cards.
    - Athlete details, verified timing/distance, and official judge commentary notes.

FR-04: STANDINGS & LEADERBOARD

- **Route:** `#/table`
- **Functional Requirements:**
  - **Interactive 3D Podium:** Animated top-3 pedestals with glowing crests, smooth number tick animations, and total medal count.
  - **Overall Medal Table:** Full breakdown of Gold, Silver, Bronze, Total Medals, Points, and 4th finishes.
  - **Category-Specific Standings View:**
    - **Football Table:** Group stage table with P, W, D, L, GF, GA, GD, PTS.
    - **Basketball Bracket:** Visual tree rendering Semi-Finals, 3rd Place Match, and Championship Final with winner paths.
  - Form guides displaying last 5 match results (W/D/L badges).

FR-05: HOUSE & TEAM PROFILES HUB

- **Routes:** `#/houses` , `#/house/:id`
- **Functional Requirements:**
  - Showcase directory of all 4 Houses (or 4 Prefectorial Teams).
  - **House Profile Page:**
    - Hero banner featuring mascot, official crest, house colors, and motto.
    - Leadership roster: House Master/Mistress, Teacher Patrons, Student House Captain.

        - Trophy showcase and historical Sports Day records.
        - Full house athlete roster with search and grade filtering.
        - Complete match history and upcoming schedule for that specific house.

FR-06: STUDENT & ATHLETE PROFILES

- **Routes:** `#/athlete/:id`, `#/student/:id`
- **Functional Requirements:**
    - Dedicated dossier for any registered student athlete.
    - Displays: Full Name, Class / Grade (e.g., JSS 2, SSS 3), Gender, Assigned House.
    - Athletic achievements: Gold, Silver, Bronze medal count, events entered, personal best times/distances.

FR-07: NEWS, MEDIA GALLERY & SOCIAL INTERACTIONS

- **Routes:** `#/news`, `#/gallery`
- **Functional Requirements:**
    - **Story Types:**
        - `video`: HTML5 player with MP4 support, video duration, and custom thumbnail poster.
        - `photo`: High-res image gallery with multi-image lightbox support.
        - `article`: Editorial journalism, match analysis, and ceremonial reports.
    - **Single-Toggle Like System:**
        - Real-time like counter persisted in Firestore (`news.likes` and `news.likedBy`).
        - Device/user key tracking in `localStorage` prevents duplicate likes; clicking again unlikes cleanly.
    - **Hashtags & Filtering:**
        - Stories tagged with athletic hashtags (e.g. `#FootballFinal`, `#100mSprint`, `#GoldMedal`).
        - Clicking any hashtag filters the active news stream; filter reset chip allows instant clearing.
    - **Live Commenting Stream:**
        - Spectators can submit real-time comments on stories.
        - Admins have permission to delete inappropriate comments.
    - **Admin Story Publisher & Editor:**
        - Verified judges can create new stories or edit existing stories (modify title, subtitle, media URL, tags with suggestion chips).

FR-08: SCRUTINEER & JUDGE SCORING CONSOLE

- **Route:** `#/admin` (gated via `requireAdmin`)
- **Functional Requirements:**
    - **Quick Event Action Table:** Live search and filter across all tournament events.
    - **Tactile Placing Form:**
        - Selection of 1st, 2nd, 3rd, and 4th place houses.
        - Dynamic athlete search input hooked to HTML5 `<datalist>` populated by 1,000+ student directory.
        - Input fields for Time (seconds), Distance (meters), or Ball Scores.
        - Official scrutineer notes field for recording record-breaking performances.
        - Instant live publishing to Cloud Firestore.
    - **Real-Time Sound Synthesizer:** Triggers procedural whistle sound on submission and celebration fanfare on gold medal allocation.

FR-09: STUDENT DATABASE & ROSTER MANAGEMENT

- **Routes:** `#/admin/students`, `#/admin/rosters`
- **Functional Requirements:**
  - Full CRUD management over student athlete directory.
  - **Search & Filter:** Instant search by student name, filter by house, filter by grade.
  - **Individual Student Operations:** Add student, edit class/gender, remove student.
  - **Bulk Import:** Paste or import multi-student datasets.
  - **Batch House Transfer Modal:** Multi-select students and transfer them en masse from one house to another.
  - **House Staff Roster Management:** Update house captain, house motto, and patron teacher lists.

FR-10: EVENT CREATOR & SCHEDULE CONFIGURATION

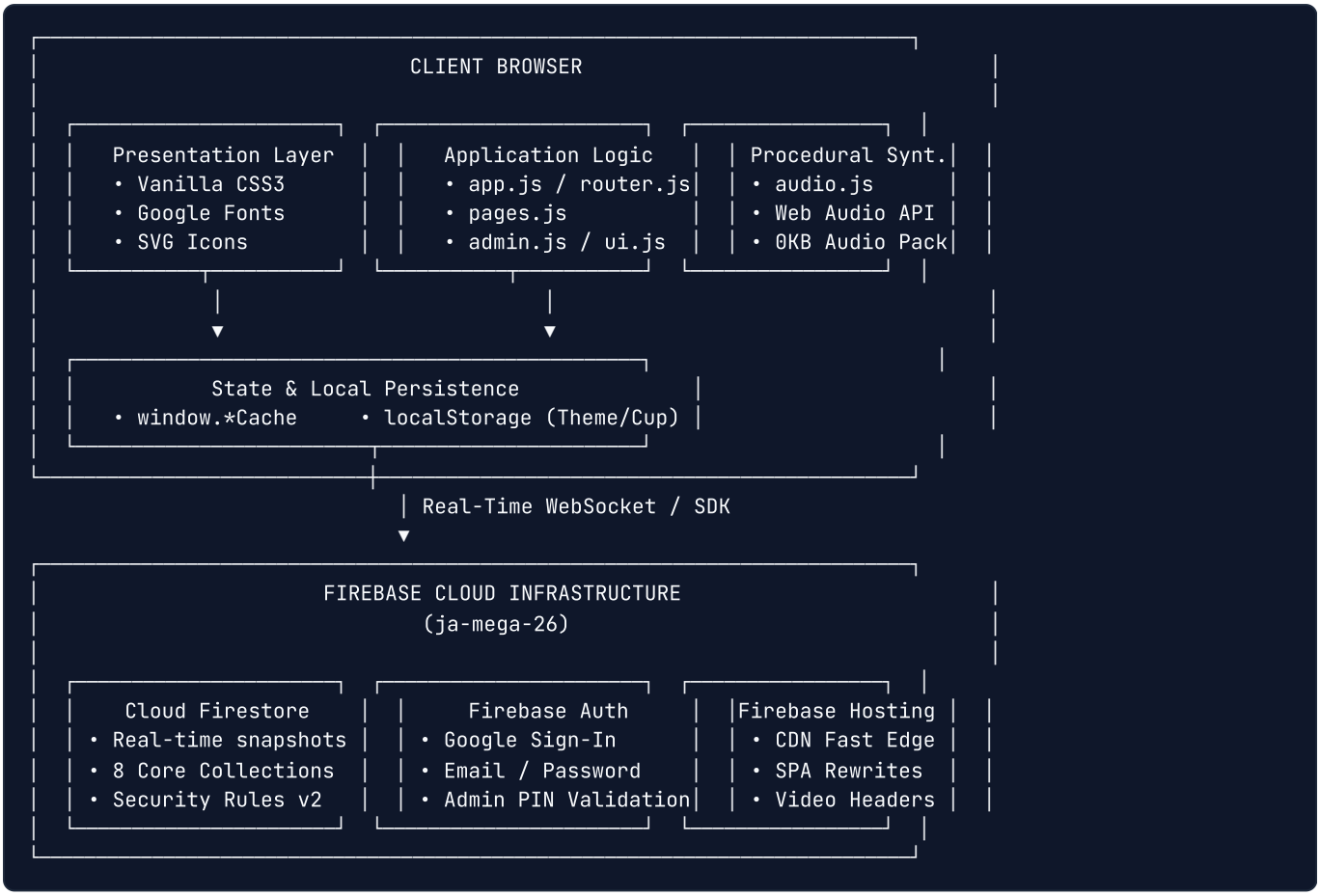
- **Routes:** `#/admin/new-event`, `#/admin/edit-event/:id`
- **Functional Requirements:**
  - Create new custom track, field, ball, or novelty events.
  - Configure event title, category, gender division, scheduled time, format (`multi` vs `h2h`), and contenders.
  - Adjust custom points matrix when applicable.
  - Edit or delete existing scheduled events.

FR-11: GOVERNANCE, OFFICIALS COUNCIL & SCRUTINEER AUDIT LOG

- **Routes:** `#/admin/admins`, `#/admin/logs`
- **Functional Requirements:**
  - **Multi-Tier Role-Based Access Control (RBAC):**
    - `super_admin`: Full administrative control, user provisioning, emergency resets.
    - `admin`: Scoring console, story publishing, event management.
    - `teacher`: Roster editing and student review.
  - **Permanent Super Admin Safeguards:**
    - Hardcoded super administrators (`ADM-001` David Festus, `ADM-003` Osaruyi Aideyan, Ihechukwu) cannot be locked out or deleted.
  - **Official PIN & Password Authentication:**
    - Direct credential validation with encrypted fallback.
  - **Immutable Scrutineer Audit Trail (`admin_audit_logs`):**
    - Every score entry, event reset, or administrative modification records: Admin ID, Admin Name, Timestamp, Action Type, Event ID, Previous Value, New Value.
  - **Emergency Sports Day Reset:**
    - Master reset action to purge scores or reload official baseline seed data, protected by a two-step confirmation dialog.

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6. Technical Architecture & Implementation Stack



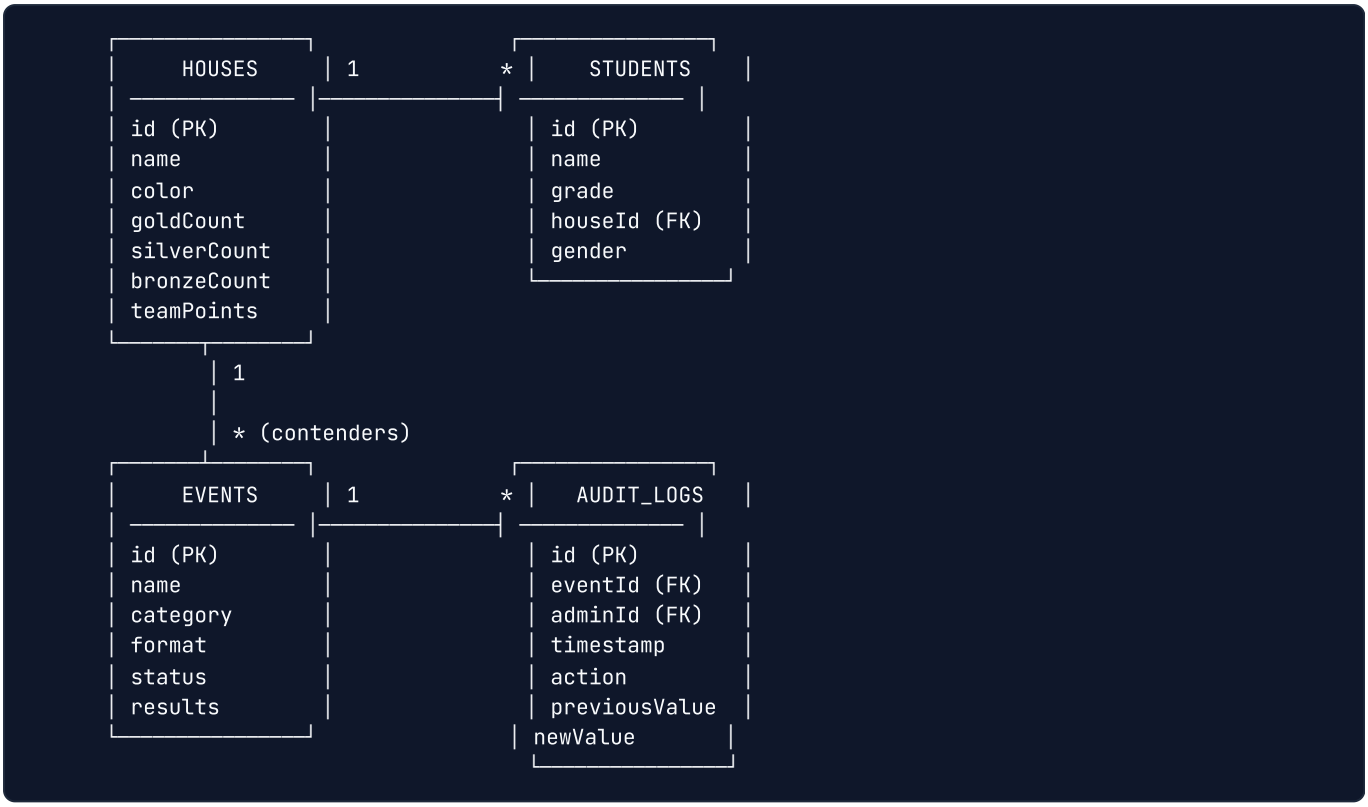
6.1 Technology Stack Details

Component	Technology	Rationale
Frontend Framework	**Vanilla ES Modules (No Framework)**	Zero build-step latency, ultra-lightweight bundle size, instant startup time on mobile devices.
Routing	Custom Client-Side Hash Router (`router.js`)	Instant screen transitions, parameter parsing (`:id`), history API integration, zero server hops.
Styling & Theming	Vanilla CSS3 + Custom Properties (`styles.css` , `app-shell.css`)	60 FPS hardware-accelerated animations, FotMob dark mode, high-contrast light mode, responsive CSS Grid.
Sound Synthesis	Web Audio API Procedural Synthesizer (`audio.js`)	Generates whistle, stadium celebration fanfare, and UI click sounds algorithmically with **0 KB external audio download**.
Realtime Database	Google Cloud Firestore (Compat SDK 10.12.0)	Multi-client real-time synchronization via `onSnapshot` listeners (< 400ms broadcast latency across stadium).
Authentication	Firebase Auth + Custom Credential Store	Google OAuth 2.0, email/password, and administrative PIN session state management.
Hosting & CDN	Firebase Hosting	Global CDN, HTTP/2 multiplexing, CORS headers for MP4 video streaming, automatic SPA wildcard rewrites.

7. Data Models & Database Schema

Firestore database instance: `(default)` in project `ja-mega-26` .

7.1 Entity Relationship Diagram



7.2 Detailed Collection Schemas

1. `HOUSES` COLLECTION

```
interface HouseDocument {
  id: string;                // "phoenix" | "eagle" | "merlin" | "swift" | "team-*"
  name: string;              // e.g. "Phoenix"
  motto: string;             // e.g. "From the Ashes, We Rise to Glory"
  mascot: string;            // e.g. "Phoenix Bird"
  color: string;             // Hex code, e.g. "#EF4444"
  colorSecondary: string;    // Hex code, e.g. "#DC2626"
  captain: string;           // e.g. "Chinedu Ihejiahi (SS 3)"
  teachers: string[];        // Array of patron names
  goldCount: number;         // Gold medals won
  silverCount: number;       // Silver medals won
  bronzeCount: number;       // Bronze medals won
  fourthCount: number;       // 4th finishes
  totalMedals: number;       // gold + silver + bronze
  teamPoints: number;        // Football + Basketball league points
  teamWins: number;          // Head-to-head match wins
  teamDraws: number;         // Head-to-head match draws
  teamLosses: number;        // Head-to-head match losses
  teamGoalsFor: number;      // Total goals / baskets scored
  teamGoalsAgainst: number;  // Total goals / baskets conceded
}
```

2. `EVENTS` COLLECTION

```
interface EventDocument {
  id: string;                 // e.g. "track-100m-boys", "pref-football-semi-1"
  name: string;               // e.g. "100m Race (Boys)"
  category: "Track" | "Field" | "Football" | "Basketball" | "Novelty" | "Ceremony";
  gender: "Boys" | "Girls" | "Mixed";
  order: number;              // Display sorting sequence
  status: "upcoming" | "in_progress" | "live" | "completed";
  format: "multi" | "h2h";    // Multi-house placement vs Head-to-Head
  scheduledTime?: string;     // e.g. "09:30"
  contenders?: string[];      // Array of House IDs, e.g. ["phoenix", "eagle"]
  knockoutRound?: "semi" | "final" | "third_place" | "group";
  results: {
    first?: { houseId: string; athlete: string; score: string; points?: number };
    second?: { houseId: string; athlete: string; score: string; points?: number };
    third?: { houseId: string; athlete: string; score: string; points?: number };
    fourth?: { houseId: string; athlete: string; score: string; points?: number };
  } | null;
  notes?: string;             // e.g. "Championship Record broken in Lane 4!"
  updatedAt?: string;         // ISO timestamp
  updatedBy?: string;         // Admin ID
}
```

3. `STUDENTS` COLLECTION

```
interface StudentDocument {
  id: string;                 // e.g. "STU-1049"
  name: string;               // e.g. "Chinedu Ihejiahi"
  grade: string;              // "JSS 1" - "SSS 3"
  gender: "Boys" | "Girls";
  houseId: string;            // "phoenix" | "eagle" | "merlin" | "swift" | "general"
  sports?: string[];          // e.g. ["100m", "4x100m", "Football"]
  goldCount?: number;
  silverCount?: number;
  bronzeCount?: number;
  personalBests?: Record<string, string>; // e.g. { "100m": "10.42s" }
}
```

4. `NEWS` COLLECTION

```
interface NewsDocument {
  id: string;                 // e.g. "news-fb-final"
  title: string;              // Headline
  subtitle?: string;          // Summary deck
  type: "video" | "photo" | "article";
  category: string;           // "Football", "Track", etc.
  mediaUrl: string;           // Asset path or external URL
  videoThumbnail?: string;    // Poster image
  videoDuration?: string;     // e.g. "1:15"
  galleryImages?: string[];   // Multi-image list for photo stories
  content: string;            // Story body
  tags: string[];             // e.g. ["FootballFinal", "GoldMedal"]
  likes: number;              // Total likes
  likedBy: string[];          // Array of voter hashes / client IDs
  comments: Array<{
    id: string;
    author: string;
    text: string;
    timestamp: string;
    isAdmin?: boolean;
  }>;
  author: string;             // e.g. "Osaruyi Aideyan (Super Admin)"
  timestamp: string;          // ISO date string
  formattedDate?: string;     // e.g. "27 Sep, 10:00"
}
```

5. `ADMINS` & `ADMIN_CREDENTIALS` COLLECTIONS

```
interface AdminDocument {
  id: string;                 // e.g. "ADM-001"
  adminId: string;
  name: string;               // e.g. "David Festus"
  email: string;              // e.g. "davidfestusugwu@gmail.com"
  title: string;              // e.g. "Chief Scrutineer"
  role: "super_admin" | "admin" | "teacher" | "official";
  createdAt: string;
}

interface AdminCredentialDocument {
  id: string;                 // email or adminId
  passwordHash: string;       // SHA-256 / PBKDF2 hash or PIN
  lastLogin: string;
}
```

6. `ADMIN_AUDIT_LOGS` COLLECTION

```
interface AuditLogDocument {
  id: string;
  timestamp: string;          // ISO timestamp
  adminId: string;            // Admin ID performing action
  adminName: string;
  action: "submit_score" | "edit_score" | "reset_scores" | "add_student" | "transfer_house";
  targetId: string;           // Event ID or Student ID
  details: string;            // Human-readable summary
  previousState?: any;
  newState?: any;
}
```

8. UI/UX Design System & Layout Specifications

8.1 Visual Aesthetic Guidelines

- **Core Inspiration:** Modern professional sports broadcast interfaces (FotMob, Sofascore, Premier League App).
- **Dark Theme (Default):**
  - App Canvas: `#0A0A0C`
  - Card Surfaces: `#121216`
  - Elevated Popups / Modals: `#181820`
  - Borders & Dividers: `#22222C`
  - Primary Text: `#FFFFFF`
  - Secondary / Muted Text: `#94A3B8`
- **Light Theme (Optional):**
  - App Canvas: `#F4F5F7`
  - Card Surfaces: `#FFFFFF`
  - Borders & Dividers: `#E2E8F0`
  - Primary Text: `#0F172A`
- **Brand Accents:**
  - Gold Trophy Accent: `#F59E0B`
  - Live Pulsing Badge: `#EF4444`
  - Completed Status: `#10B981`

8.2 Typography

- **Headings & Brand Titles:** `Outfit` (Weights: 600, 700, 800, 900) via Google Fonts. High-energy, athletic, modern.
- **Data Tables & Body Text:** `Inter` (Weights: 400, 500, 600, 700). Ultra-crisp numerical tabular alignment and readability.

8.3 Responsive Layout Strategy

DESKTOP (SCREEN WIDTH >= 1024PX)

- **Global Sticky Header:** Brand crest, tournament cup switcher badge, quick global search bar, desktop text navigation links, theme toggle, official login portal button.
- **3-Column Grid Shell:**
  - **Left Sidebar (280px):** Live House standings ranking widget with miniature crests and points, plus category quick filters.
  - **Center Viewport (Flex 1):** Main active page screen (Matches, Standings Table, Event Details, Admin Console).
  - **Right Sidebar (340px):** Real-time News, video highlight reel player, and latest track results feed.

MOBILE & TABLET (SCREEN WIDTH < 1024PX)

- **Single-Column App Shell:** Compact header with current tournament badge and theme toggle.
- **Sticky Bottom Navigation Bar (5 Tabs):**
  1. `Matches` (`#/events`) — Icon: Calendar
  2. `Table` (`#/table`) — Icon: Trophy
  3. `News` (`#/news`) — Icon: Newspaper
  4. `Houses` (`#/houses`) — Icon: Shield / House
  5. `Judge` (`#/admin`) — Icon: Whistle / Gavel *(Conditionally rendered for authenticated officials only)*

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9. Security, Governance & Audit Architecture

9.1 Firestore Security Rules (`firestore.rules`)

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    function isAuthenticated() {
      return request.auth != null;
    }

    // Public live read access for tournament transparency
    match /houses/{houseId} {
      allow read: if true;
      allow write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }

    match /events/{eventId} {
      allow read: if true;
      allow write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }

    match /students/{studentId} {
      allow read: if true;
      allow write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }

    match /news/{newsId} {
      allow read: if true;
      allow write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }

    match /admin_audit_logs/{logId} {
      allow read, write: if isAuthenticated() || request.time < timestamp.date(2026, 10, 22);
    }

    match /admins/{adminId} {
      allow read, write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }

    match /admin_credentials/{credId} {
      allow read, write: if request.time < timestamp.date(2026, 10, 22) || isAuthenticated();
    }
  }
}
```

9.2 Immutable Audit Trail & Dispute Resolution

- Every mutation to an event score generates an audit log entry in `/admin_audit_logs`.
- Logs record the authenticated official's ID, name, precise timestamp, before/after score payload, and reason.
- Guarantees transparency and eliminates scoring disputes between house masters.

---

10. Quality Assurance & Verification Matrix

The application codebase contains an extensive automated and end-to-end verification suite in the `scratch/` directory:

Test Script	Verification Focus	Expected Output	Status
`verify_all_requirements.js`	End-to-end verification of News like/unlike, tag filtering, story editor, football group table, and basketball bracket.	All 6 test suites pass; generates screenshot artifacts.	**Verified**
`verify_cup_switch_e2e.js`	Real-time tournament switcher between Interhouse and Prefectship Cup.	State isolation confirmed; correct house rosters rendered.	**Verified**
`verify_points_removal.js`	Ensures track/field events strictly award Olympic medals without legacy points.	Medal ranking mathematically verified.	**Verified**
`simulate_full_championship.mjs`	Full simulation of 35 championship events with realistic race times and football scores.	Firestore batch updates succeed; standings recalculate accurately.	**Verified**
`verify_h2h_scores.js`	Football head-to-head score parsing, drawn match handling, and league points award.	Win (3 pts), Draw (1 pt), Loss (0 pts) confirmed.	**Verified**
`test_athlete_stat_matching.js`	Links student registry with placing results and verifies athlete profile stats.	Medals correctly aggregated per student.	**Verified**
`verify_production_osaruyi.js`	Super admin session validation and permissions check.	Full super admin capabilities confirmed.	**Verified**

11. Deployment, Hosting & Operations Runbook

11.1 Firebase Hosting Configuration (`firebase.json`)

```
{
  "hosting": {
    "public": "public",
    "ignore": ["firebase.json", "**/.*", "**/node_modules/**"],
    "headers": [
      {
        "source": "**/*.@(mp4|webm)",
        "headers": [
          { "key": "Access-Control-Allow-Origin", "value": "*" },
          { "key": "Cache-Control", "value": "public, max-age=604800" }
        ]
      }
    ],
    "rewrites": [
      { "source": "**", "destination": "/index.html" }
    ]
  }
}
```

11.2 Deployment Commands

- **Deploy Full Stack:**

  ```
  firebase deploy --project ja-mega-26
  ```

- **Deploy Hosting Only:**

```
firebase deploy --only hosting --project ja-mega-26
```

- **Deploy Firestore Security Rules & Indexes:**

```
firebase deploy --only firestore --project ja-mega-26
```

11.3 Local Development Server

- To run locally without build dependencies:

```
python scratch/dev_server.py 5055
# or
npx -y serve public -l 5055
```

- Access in browser: `http://localhost:5055`
- Super Admin autologin test query: `http://localhost:5055/?admin=osaruyi` or `http://localhost:5055/?autologin=admin`

---

12. Future Roadmap & Enhancements

1. **Push Notifications:** Web Push API integration for instant goal, race start, and gold medal alert notifications to spectators.
2. **Live Video Streaming (HLS/WebRTC):** Embedding low-latency stadium camera feeds directly into the Event Detail and News header.
3. **Parent & Alumni March Past Voting:** Interactive spectator voting for the Best Dressed House and Most Creative March Past Salute.
4. **Automated Certificate & Official Results Export:** One-click PDF generation of official event result sheets signed by the Chief Scrutineer and Head of School.
5. **Wearable / Smart Stopwatch BLE Integration:** Direct Bluetooth Low Energy link with digital stopwatches to feed sprint times directly into the scoring console.
